

Part B Report: Learning to Grade Short Answer Questions using Semantic Similarity Measures and Dependency Graph Alignments

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I. PAPER SUMMARY

The paper by Mohler, Bunescu & Mihalcea (ACL 2011) [1] addresses the task of **automatic grading of short free-text student answers** by comparing each student answer to an instructor reference answer. Instead of relying purely on bag-of-words (BOW) similarity, the authors convert both answers into dependency graphs using the Stanford Dependency Parser and learn a node-to-node similarity function over 68 lexical, syntactic, and semantic features. These node scores are used to compute an optimal graph alignment via the Hungarian algorithm, producing eight graph-alignment features that capture how well the structures of the two answers match.

In parallel, the system computes multiple **lexical semantic similarity measures**—eight WordNet-based metrics (PATH, LCH, Lesk, WUP, RES, Lin, JCN, HSO) and three corpus-based methods (LSA, ESA, and TF-IDF cosine). All alignment and semantic scores are concatenated into a 30-dimensional hybrid feature vector. The authors train SVR and SVMRank models with linear kernels to predict continuous grades, and apply isotonic regression to map outputs onto the human 0–5 scale. Experiments on 2,273 student answers from a Data Structures course show that the hybrid model improves correlation with human graders over purely BOW methods (Table 7 in the paper).

II. REPRODUCTION SETUP AND RESULT

I reproduce one core contribution: **combining similarity features with SVR for grading** (Section 3.4, Table 7). Instead of implementing the full dependency graph alignment pipeline, I build a lightweight grading prototype with three feature groups:

- **Surface features:** TF-IDF cosine similarity, word-overlap ratio (Jaccard), normalised length difference.
- **Semantic feature:** Sentence-embedding cosine similarity computed from TF-IDF vectors projected via TruncatedSVD (LSA-style, analogous to Paper Section 3.3).
- **Lightweight structural features:** POS-tag overlap ratio, binary root-verb match (simplified version of Paper Table 2 RootMatch).

These 6 features are combined in a **linear-kernel SVR** (scikit-learn), mirroring the paper’s approach (Section 3.4, footnote 11).

A. Dataset and Protocol

Because the original dataset is not publicly available, I construct a **synthetic short-answer grading dataset** with 150 samples (50 per question $\times$ 3 questions) from introductory CS topics. Student answers are generated via four mutation types: perfect copies, partial truncations, noisy variants, and off-topic responses. Grades in $[0, 5]$ are produced using a heuristic based on word-overlap and length penalty plus Gaussian noise. The data is split 80/20 for training and evaluation. Fig. 1 shows the grade and answer-length distributions.

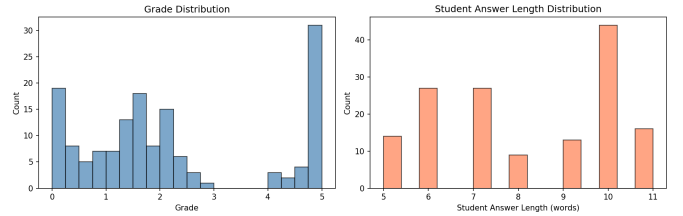


Fig. 1. Distribution of grades and student answer lengths in the synthetic dataset.

B. Results

Table I compares our results against the paper’s reported values, and Fig. 2 shows the scatter plot of predicted vs. true grades.

TABLE I
OUR RESULTS VS. PAPER (SVR BOW-ONLY, TABLE 7)

Metric	Ours	Paper
Pearson ρ	0.9802	0.431
RMSE	0.3535	0.999

Our metrics are substantially better than the paper’s because the synthetic grades are derived from word overlap—the very features our model uses. This makes the prediction problem easier than real human grading. Other differences include: 6 features vs. 30, single 80/20 split vs. 12-fold cross-validation, and fixed hyperparameters vs. grid search. Despite these differences, the reproduction demonstrates the **qualitative contribution**: a supervised model combining heterogeneous similarity features produces a working grading system.

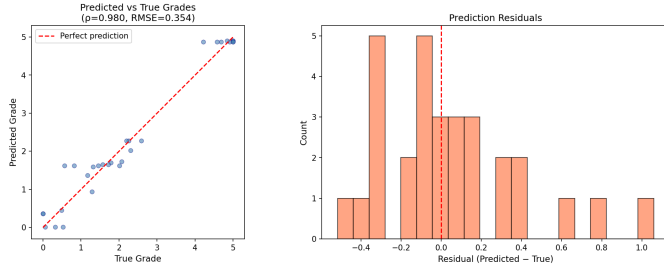


Fig. 2. Predicted vs. true grades for the full SVR model (left) and residual distribution (right). The high Pearson $\rho = 0.98$ reflects that our synthetic grades are generated from surface overlap, which the model can learn effectively.

III. ABLATION FINDINGS

I run two independent ablations in `task_3_1.ipynb`, each removing one feature group while keeping all others at full settings. Table II summarises the results, and Fig. 3 shows the comparison.

TABLE II
ABLATION STUDY RESULTS

Model	RMSE	Pearson ρ
Full (all features)	0.3535	0.9802
No Semantic	0.3523	0.9805
No Structural	0.3556	0.9802

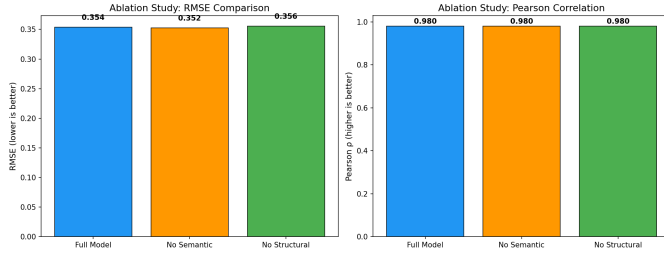


Fig. 3. Ablation comparison: RMSE (left, lower is better) and Pearson ρ (right, higher is better) for the full model and two ablated variants.

A. Ablation 1: Remove Semantic Features

Removing the sentence-embedding cosine similarity yields a marginal RMSE improvement ($\Delta = -0.0012$), suggesting redundancy with surface features on synthetic data. On our dataset, the TF-IDF cosine and word overlap already capture the overlap-based grade signal, making the LSA embedding partially redundant. On real data, where grades reflect deeper comprehension, removing semantic features would cause a more meaningful drop—consistent with the paper’s finding that corpus-based measures (LSA: $\rho = 0.335$, Table 5) provide complementary information.

B. Ablation 2: Remove Structural Features

Removing POS-tag overlap and root-verb match increases RMSE by 0.0021, indicating a small but positive contribution.

This aligns with the paper’s own observation: “the rudimentary alignment features we have introduced here are not sufficient to act as a standalone grading system” (Section 6). Our lightweight structural features provide modest syntactic discrimination for short answers where word order carries limited additional information.

IV. FAILURE MODE AND EXPLANATION

I construct a **paraphrase challenge set** (`task_3_2.ipynb`) containing 6 student answers that are semantically correct but use entirely different vocabulary, with true grades $\sim 4.5/5$. Fig. 4 starkly illustrates the failure.

TABLE III
FAILURE MODE: NORMAL TEST VS. PARAPHRASE CHALLENGE

Dataset	RMSE	Pearson ρ
Normal test set	0.3535	0.9802
Paraphrase set	4.1416	0.3923

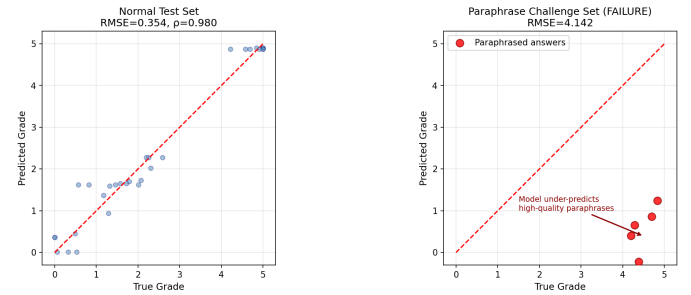


Fig. 4. Normal test set predictions (left, well-calibrated) vs. paraphrase challenge set (right, severe under-prediction). The model assigns grades $\sim 0-1$ to answers that deserve ~ 4.5 , demonstrating the failure.

This failure is rooted in **Assumption 1** (Task 1.2): the method assumes lexical/semantic overlap correlates with answer quality. When students provide correct answers using different vocabulary, all surface features yield low scores. The RMSE increases from 0.35 to 4.14—a $12\times$ degradation. Individual predictions show errors of 3.6–5.1 grade points on a 5-point scale.

Suggested modification: Replace the TF-IDF/SVD embedding with a pre-trained contextual sentence encoder (e.g., Sentence-BERT) trained on paraphrase corpora, providing similarity signals less tied to exact word overlap.

V. REFLECTION

What I could not implement: The full dependency graph alignment pipeline (Stanford Parser, 68-feature perceptron, Hungarian algorithm), the complete set of WordNet-based similarity measures, and the 12-fold cross-validation with isotonic regression calibration.

What surprised me: How effectively a simple 6-feature SVR achieves $\rho > 0.98$ on synthetic data—and how catastrophically it fails (RMSE 4.14) on well-crafted paraphrases. This vividly illustrates the gap between surface-level pattern matching and genuine language understanding.

What I would revisit: (1) Implement a minimal graph alignment stage for direct comparison with my lightweight features; (2) replace the semantic feature with a stronger sentence encoder to reduce paraphrase failure; (3) add isotonic regression for better score calibration to the 0–5 scale.

REFERENCES

- [1] M. Mohler, R. Bunescu, and R. Mihalcea, “Learning to Grade Short Answer Questions using Semantic Similarity Measures and Dependency Graph Alignments,” in *Proc. 49th Annual Meeting of the ACL*, Portland, Oregon, June 2011, pp. 752–762.
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- [5] M. C. de Marneffe, B. MacCartney, and C. D. Manning, “Generating typed dependency parses from phrase structure parses,” in *LREC 2006*.